

Apurva Mandalika

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SUMMARY

AI/ML Engineer with 5+ years of experience building ML and applied AI solutions across e-commerce, hospitality, retail, and customer support. Experienced in Python, SQL, PyTorch, APIs, and production-style AI systems, with recent work in RAG, hybrid retrieval, reranking, and forecasting copilots. Recognized with 4 performance awards, a hackathon win, and 2 promotions in three years.

SKILLS

ML/AI Frameworks: PyTorch, Scikit-learn, Sentence Transformers; **LLM / AI Systems:** RAG, Prompt Engineering, Hybrid Retrieval, BM25, Reranking, FAISS, Gemini API; **Languages:** Python, SQL, Java, JavaScript, Swift, Ruby; **Data Analytics & Visualization:** Pandas, NumPy, Tableau, Matplotlib; **Software Development:** FastAPI, Flask, REST APIs, Streamlit, React.js, Docker, Git, GitHub, API Integration; **Cloud Technologies:** AWS, GCP.

EXPERIENCE

Founding AI Engineer | *RetailIQ AI (Self-Initiated Project)*

Jan 2026 – Present

- Built a production-style retail knowledge assistant using Retrieval-Augmented Generation to answer questions from pricing, promotions, returns, inventory, and store operations documents with grounded source citations.
- Implemented a hybrid retrieval pipeline using FAISS semantic search, BM25 keyword retrieval, and cross-encoder reranking to improve relevance for policy-heavy and business-process queries.
- Integrated Gemini for answer generation and designed metadata-aware chunking, query logging, and FastAPI-based backend workflows to support more reliable retrieval and observability.
- Developed an evaluation dataset and retrieval pipeline that achieved 100% source retrieval coverage and ~89% answer reference-point coverage across 15+ retail QA evaluation queries.

AI/ML Senior Consultant | *Heartland Community Network*

Sep 2025 – Present

- Unified inventory data across a client's physical boutique and Shopify storefront using Python, eliminating 300+ duplicate SKUs and reducing stock discrepancies by 94%.
- Implemented a customer segmentation pipeline for a retail client using K-Means clustering, identifying 4 distinct buyer personas and increasing targeted campaign ROI by 22%.

AI/ML | *[24]7.ai*

Jan 2019 – Jul 2023

- Introduced and implemented ARKit- and computer-vision-based workflows for AR video-based customer support solutions on iOS and Web, improving issue resolution rates by 50% compared to traditional methods.
- Contributed 15+ POCs of AR features, securing a patent with USPTO - PCT/IB2023/050635.
- Built and maintained a model performance dashboard in Excel using Hive, SQL, and Python to track KPIs for 23 client predictive targeting models, increasing actionable insights by 40%.
- Created Time on Page (TOP) and Page-Level Propensity to Purchase after Chat (P2PC) models, increasing chat propensity by 8%, RPC by 6%, and conversion rates by 12%.
- Deployed scalable predictive targeting models for 4 clients using Hadoop and FlashML for customer behavior modeling.
- Improved maintainability of legacy and deployed models by creating structured JIRA/Confluence documentation, while consolidating analytics and reporting workflows.

PROJECTS

Walmart Demand Forecasting Copilot | *Python, Forecasting, LLM Routing, FastAPI, XGBoost, Streamlit*

- Developed an AI-powered retail analytics assistant that uses Gemini-based routing to interpret natural language queries and dynamically invoke forecasting or KPI tools enabling a lightweight agent-style workflow for demand planning.
- Built and served a weekly demand forecasting pipeline with XGBoost, lag features, and rolling-window features, integrating predictions and KPI summaries into a FastAPI + Streamlit application for interactive analytics.

Heart Disease Risk Prediction API | *Python, Scikit-learn, Flask, Docker*

Trained and optimized a heart-disease classification model (F1: 0.82, AUC: 0.89). Architected a production-ready API for real-time risk prediction with automated client-side testing and containerized the deployment for seamless portability.

Deep Learning Model for Image Classification | *Python, Jupyter Notebook, PyTorch*

Designed a hybrid deep learning model combining DenseNet and ResNet architectures for CIFAR-10 image classification, showcasing model architecture optimization, feature abstraction, and training automation. Achieved an accuracy of 92.5%.

Multimodal Classification Model | *Python, Jupyter Notebook, PyTorch*

Developed a fusion model combining a CNN for image data and an ANN for audio data to classify the multimodal MNIST dataset and achieved a validation accuracy of 98.92%. Emphasized representation learning and input heterogeneity through this model.

EDUCATION

Texas A&M University

Master's in Computer Science

Amrita Vishwa Vidyapeetham

Bachelor of Technology, Computer Science and Engineering

College Station, TX

Aug 2023 – May 2025

Amritapuri, Kerala

Aug 2015 – May 2019